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STW4005CEM: Database System

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Abstract

This documentation report explores the transformation of a paper-based database into a digitalized relational database management system(RDBMS). Moreover, this report also includes data visualization using popular Python Libraries such as Numpy, Pandas, and Matplotlib.

Keywords



database
numpy
normalization
rdbms
er-diagram matplotlib
pandas

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Introduction

In such a digitalized world, managing a database and records manually on paper is a tough task ([Data Management in the Era of Digitalization, 2019](#)). This project aims to solve this issue of the complete transformation of the records of Digi Medical HealthCare into a managed database.

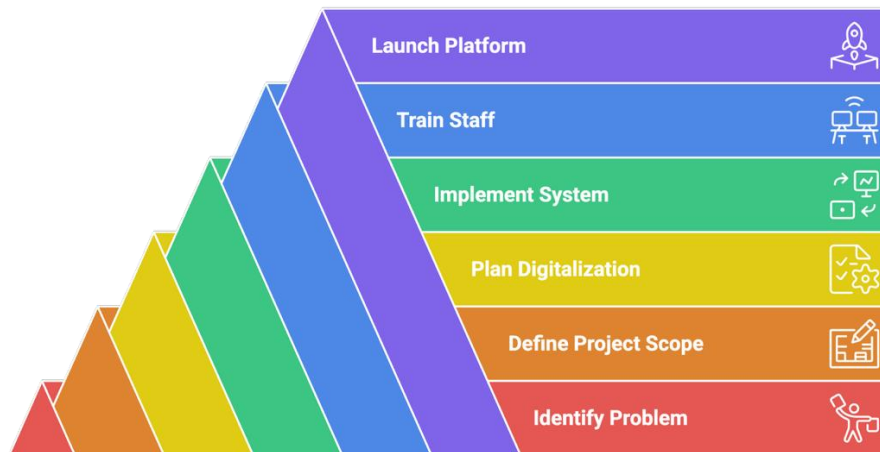


Figure 1: Steps to Transform a Paper-Based Database to Digital Database

Specifically, four different sections are divided for this complete transformation:

1. Normalization
2. Entity-Relationship Diagram
3. SQL Implementation
4. Data Visualization

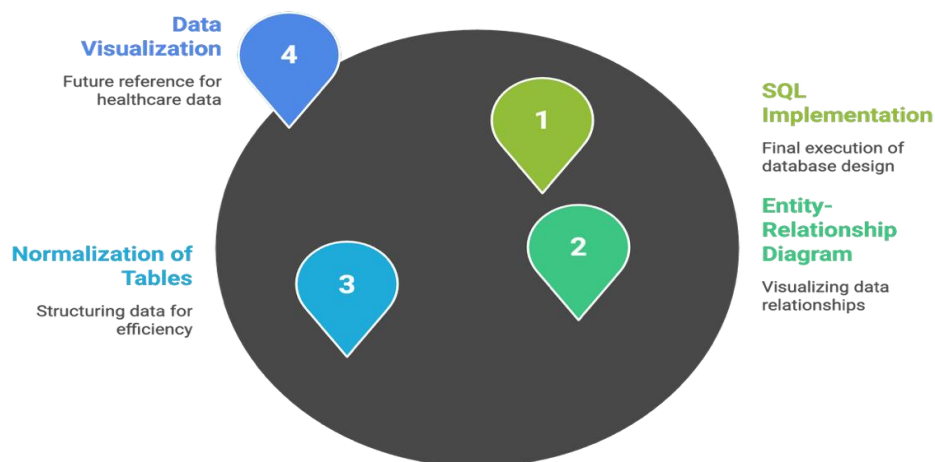


Figure 2: Four Strategies for Digital Database Transformation

Task 1: Design

Normalization Process

Normalization helps to organize the tables and data by minimizing redundancy and data anomalies ([Data Normalization, n.d.](#)). For this project, tables are normalized upto third normal form(3NF).

Table 1: Party Details

Host Ref	Name	Date	Time	No of Guests
12	Krishna Rai	12/01/2018	8pm	23
14	Jenisha Sharma	24/03/2018	7.30pm	11
34	Kritika Thapa	08/04/2018	11.00am	8
67	Anish Singh	25/04/2108	3pm	12

Figure 3: Original Party Details Table

1NF is already achieved because all cells are atomic.

Host Ref	Name	PartyID	Host Ref	Date	Time	No of Guests
12	Krishna Rai	1	12	12/01/2018	8pm	23
14	Jenisha Sharma	2	14	24/03/2018	7.30pm	11
34	Kritika Thapa	3	34	08/04/2018	11.00am	8
67	Anish Singh	4	67	25/04/2108	3pm	12

Figure 4: Host_Ref and Party_Data

However, not all fields depend on HostRef, so the separation gives 2NF.

Host Ref
12
14
34
67

Figure 5: Host_Ref Table

HostRef can be used directly as primary/foreign key, achieving 3NF.

Table 2: Customer Details

S.N	Name	Address	Number	Email	Customer Type
12	Krish Rai	Jhapa	9845641792	krish@hmail.com	Host
2	Anish Kc	Chitwan	9844532468	anish@gmail.com	Customer
3	Om Thapa	Pokhara	9845431792	om12@gmail.com	Ambassador
4	Raj Joshi	Butwal	9845124793	raj@gmail.com	Customer

Figure 6: Original Customer Details Table

This table is already in 3NF Form (i.e: no atomic values, no functional dependencies, and no transitive dependencies)

Table 3: Order Details

Order Ref	Customer Ref	Payment Method	Product Code	Description	Price	Qty	Total Price	Type	Future Host?
1	3	Credit Card	123	ABC + free eye refresh	52	25	1300	Email	N
2	4	Cash	23	Travel Essentials	30	1	30	Party	Y
3	2	Cheque	235	Super Deluxe Collection	35	1	35	Party	N
4	3	Cash	23	Travel Essentials	30	5	150	Email	N

Figure 7: Original Order Details Table

1NF is already achieved because all cells are atomic.

Now there is a functional dependency which must be removed to achieve 2NF.

Order Ref	Customer Ref	Payment Method	Qty	Total Price	Type	Future Host
1	3	Credit Card	123	1300	Email	N
2	4	Cash	23	30	Party	Y
3	2	Cheque	235	35	Party	N
4	3	Cash	23	150	Email	N

Order Ref	Product Code	Description	Price
1	123	ABC + free eye refresh	52
2	23	Travel Essentials	30
3	235	Super Deluxe Collec	35
4	23	Travel Essentials	30

Figure 8: Order_Data Second Normal Form

Now there is a transitive dependency which should be removed to achieve 3NF.

Order Ref	Customer Ref	Payment Method	Type	Future Host
1	3	Credit Card	Email	N
2	4	Cash	Party	Y
3	2	Cheque	Party	N
4	3	Cash	Email	N

Product Code	Description	Price
123	ABC + free eye refres	52
23	Travel Essentials	30
235	Super Deluxe Collecti	35

Trace_ID	Order Ref(FK)	Product Code(FK)	Qty	Total Price
1000	1	123	25	1300
1001	2	23	1	30
1002	3	235	1	35
1003	4	23	5	150

Figure 9: Order_Data,Product_Data,Order_Tracking Tables

Table 4: Customer Type

Type	Comments
Customer	Standard price
Host	-2%
Ambassador	15% of profit

Figure 10: Original Customer Type Table

This table is already in 3NF Form.

Table 5: Delivery Information

Order Ref	Customer Name	Total Cost	Discount Awarded?
1	Krish Rai	1450	Yes
2	Anish Kc	30	No
3	Om Thapa	35	Yes

Figure 11: Original Delivery Information Table

This table is already in 3NF Form.

Order Ref	CustomerRef	Total Cost	Discount Awarded?
1	12	1450	Yes
2	2	30	No
3	3	35	Yes

Figure 12: Normalized Delivery Information Table

Table 6: Product Information

Product Code	Name	Description	Price	Supplier	Category
123	ABC + free eye refresh	Cleanser- Natural Ingredients	Rs.52.00	Righthhealth	Skincare
23	Travel Essentials	Shampoo and Conditioner	Rs.30	Royams	Haircare
234	Smooth Skin Collection	Skincare Products – Cleanser, Toner, Moisturiser	Rs.45	Sincere Ltd	Skincare
235	Super-Deluxe Collection	Shower Collection	Rs.35	Grenfell UK	Bodycare

Figure 13: Original Product Information Table

1NF is already achieved because all cells are atomic.

Now there is a functional dependency which must be removed to achieve 2NF.

Product Code	Name	Description	Price	SupplierID(FK)	CategoryID(FK)
123	ABC + free eye refresh	Cleanser- Natural Ingredients	Rs.52.00	35	1
23	Travel Essentials	Shampoo and Conditioner	Rs.30	38	4
234	Smooth Skin Collection	Skincare Products – Cleanser, Toner, Moisturiser	Rs.45	39	1
234	Super-Deluxe Collection	Shower Collection	Rs.35	45	2

Figure 14: Product Information Normalized Table

This table does not consist any transitive dependency. Hence, it is in 3NF.

Table 7: Category Information

Category ID	Category Name	Comments
1	Skincare	Summer Season
2	Bodycare	Summer Range
3	Various	All minerals
4	Haircare	All products

Figure 15: Original Category Information Table

This table is already in 3NF Form.

Table 8: Supplier Information

Supplier ID	Name	Email	Tel Number
12	ABC	supplier@abc.net	1282456762
1	CDE	supplier@cde.co	1252776789
34	DEF	supplier@def.com	1705674890
2	GHI	supplier@ghi.com.np	1200467876

Figure 16: Original Supplier Information Table

This table is already in 3NF Form.

Entity-Relationship for Digi Medical HealthCare

This Entity-Relationship Diagram describes major entities such as:

1. Customer_Type
2. Host_Data
3. Party_Data
4. Customer_Details
5. Delivery_Information
6. Supplier_Data
7. Order_Data
8. Order_Tracking
9. Product_Data
10. Category_Data

Moreover, foreign key and primary key are used in the ER Diagram to show more relation between them([GeeksforGeeks, 2025](#)).

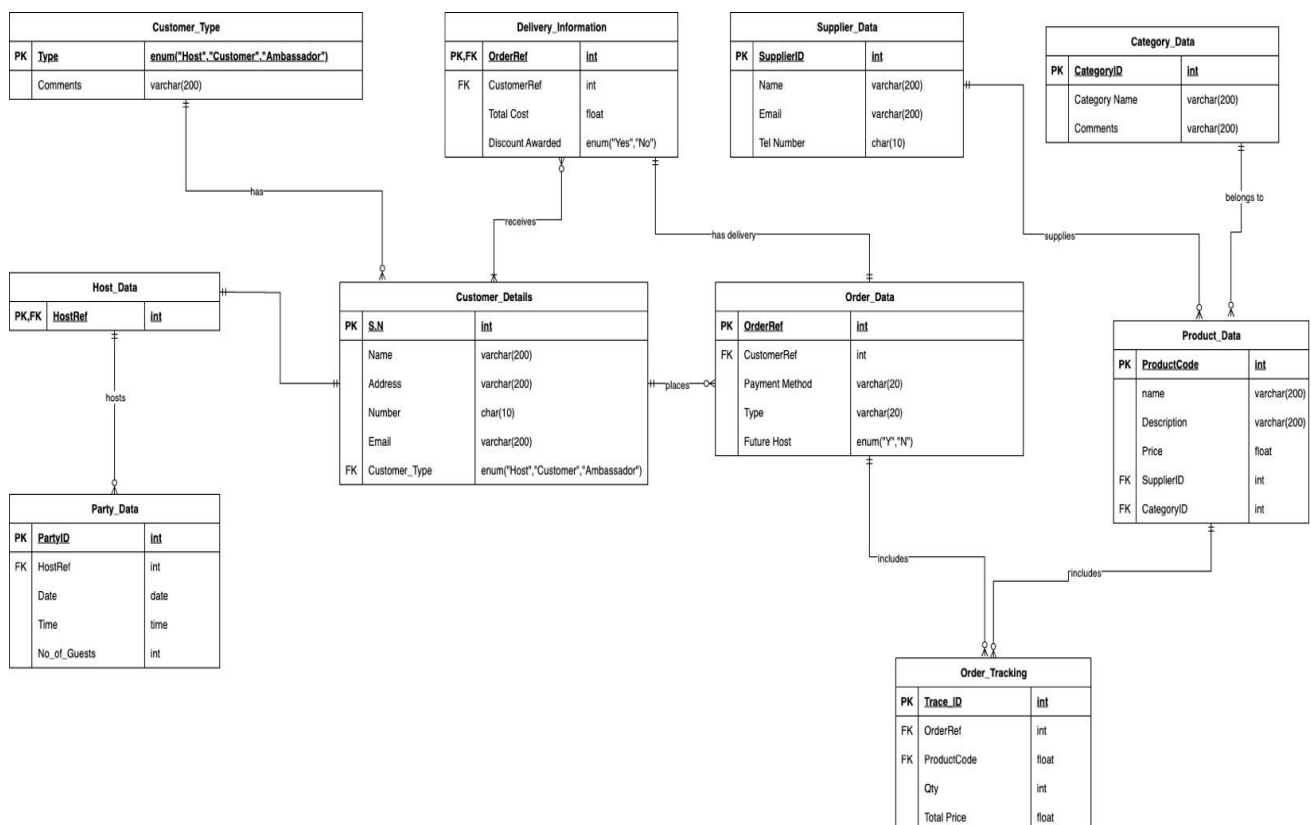


Figure 17: ER Diagram of Digi Medical Health Care

Task 2: SQL Implementation

MYSQL Workbench is used as a tool for these SQL queries.

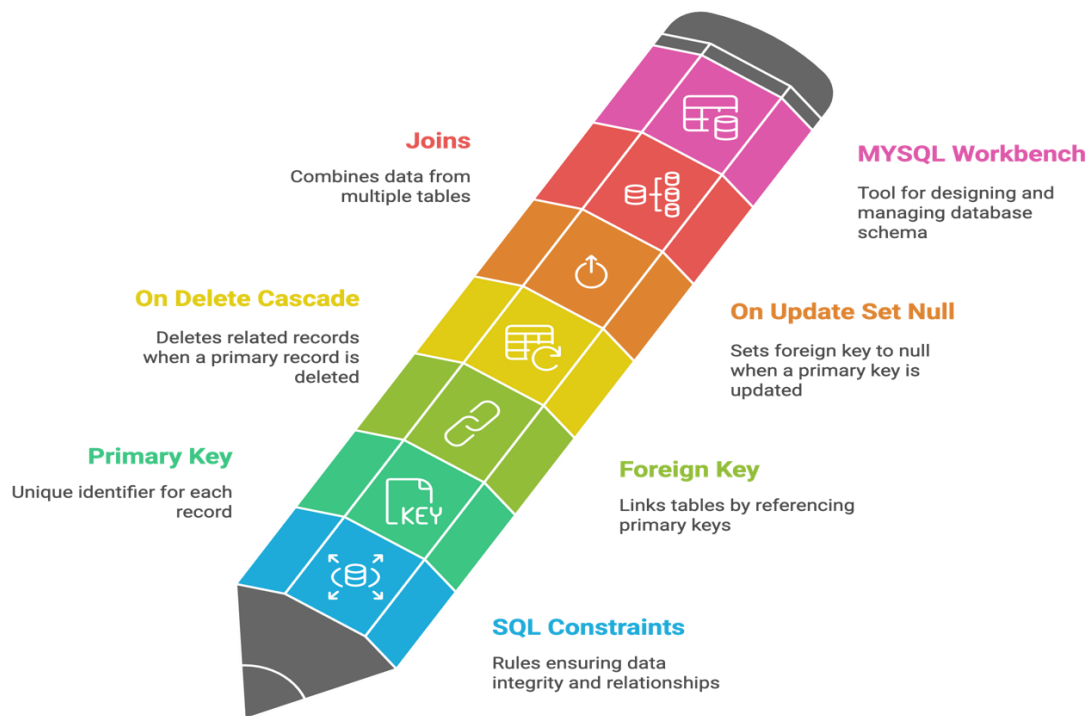


Figure 18: Design Preview

Table Creation Query

```

7 • CREATE TABLE Customer_Type (
8     Type ENUM('Host', 'Customer', 'Ambassador') PRIMARY KEY,
9     Comments VARCHAR(200)
10 );
11
12 • CREATE TABLE Customer_Details (
13     SN INT PRIMARY KEY,
14     Name VARCHAR(200),
15     Address VARCHAR(200),
16     Number CHAR(10),
17     Email VARCHAR(200),
18     Customer_Type ENUM('Host', 'Customer', 'Ambassador'),
19     FOREIGN KEY (Customer_Type) REFERENCES Customer_Type(Type) ON DELETE CASCADE
20 );
21
22 • CREATE TABLE Host_Data (
23     HostRef INT PRIMARY KEY,
24     FOREIGN KEY (HostRef) REFERENCES Customer_Details(SN) ON DELETE CASCADE
25 );

```

```

27 • CREATE TABLE Party_Data (
28     PartyID INT PRIMARY KEY,
29     HostRef INT,
30     Date DATE,
31     Time TIME,
32     No_of_Guests INT,
33     FOREIGN KEY (HostRef) REFERENCES Host_Data(HostRef) ON DELETE CASCADE
34 );
35
36 • CREATE TABLE Supplier_Data (
37     SupplierID INT PRIMARY KEY,
38     Name VARCHAR(200),
39     Email VARCHAR(200),
40     Tel_Number VARCHAR(20)
41 );
42
43 • CREATE TABLE Category_Data (
44     CategoryID INT PRIMARY KEY,
45     Category_Name VARCHAR(200),
46     Comments VARCHAR(200)
47 );
48
49 • CREATE TABLE Product_Data (
50     ProductCode INT PRIMARY KEY,
51     name VARCHAR(200),
52     Description VARCHAR(200),
53     Price FLOAT,
54     SupplierID INT,
55     CategoryID INT,
56     FOREIGN KEY (SupplierID) REFERENCES Supplier_Data(SupplierID) ON DELETE CASCADE,
57     FOREIGN KEY (CategoryID) REFERENCES Category_Data(CategoryID) ON DELETE CASCADE
58 );
59
60 • CREATE TABLE Order_Data (
61     OrderRef INT PRIMARY KEY,
62     CustomerRef INT,
63     Payment_Method VARCHAR(20),
64     Type VARCHAR(20),
65     Future_Host ENUM('Y', 'N'),
66     FOREIGN KEY (CustomerRef) REFERENCES Customer_Details(SN) ON DELETE CASCADE
67 );
68

```

```

69 • CREATE TABLE Order_Tracking (
70     Trace_ID INT PRIMARY KEY,
71     OrderRef INT,
72     ProductCode INT,
73     Qty INT,
74     Total_Price FLOAT,
75     FOREIGN KEY (OrderRef) REFERENCES Order_Data(OrderRef) ON DELETE CASCADE,
76     FOREIGN KEY (ProductCode) REFERENCES Product_Data(ProductCode) ON DELETE CASCADE
77 );
78
79 • CREATE TABLE Delivery_Information (
80     OrderRef INT,
81     CustomerRef INT,
82     Total_Cost FLOAT,
83     Discount_Awarded ENUM('Yes', 'No'),
84     PRIMARY KEY (OrderRef, CustomerRef),
85     FOREIGN KEY (OrderRef) REFERENCES Order_Data(OrderRef) ON DELETE CASCADE,
86     FOREIGN KEY (CustomerRef) REFERENCES Customer_Details(SN) ON DELETE CASCADE
87 );
88

```

Figure 19: Table Creation Query

Data Insertion Query

```

92 -- Customer Types
93 INSERT INTO Customer_Type VALUES
94     ('Host', '-2%'),
95     ('Customer', 'Standard price'),
96     ('Ambassador', '15% of profit');
97
98 -- Customer Details
99 INSERT INTO Customer_Details VALUES
100     (12, 'Krish Rai', 'Jhapa', '9845641792', 'krish@hmail.com', 'Host'),
101     (2, 'Anish Kc', 'Chitwan', '9844532468', 'anish@gmail.com', 'Customer'),
102     (3, 'Om Thapa', 'Pokhara', '9845431792', 'om12@cmail.com', 'Ambassador'),
103     (4, 'Raj Joshi', 'Butwal', '9845124793', 'raj@smail.com', 'Customer');
104
105 -- Hosts
106 INSERT INTO Host_Data VALUES (12);
107
108 -- Parties
109 • INSERT INTO Party_Data VALUES
110     (1, 12, '2018-01-12', '20:00:00', 23);
111
112 -- Suppliers
113 INSERT INTO Supplier_Data VALUES
114     (12, 'ABC', 'supplier@abc.net', '1282456762'),
115     (1, 'CDE', 'supplier@cde.co', '1252776789'),
116     (34, 'DEF', 'supplier@def.com', '1705674890'),
117     (2, 'GHI', 'supplier@ghi.com.np', '1200467876');
118

```

```

119  -- Categories
120  INSERT INTO Category_Data VALUES
121      (1, 'Skincare', 'Summer Season'),
122      (2, 'Bodycare', 'Summer Range'),
123      (3, 'Makeup', 'All minerals'),
124      (4, 'Haircare', 'All products');
125
126  -- Products
127  INSERT INTO Product_Data VALUES
128      (123, 'ABC + free eye refresh', 'Cleanser- Natural Ingredients', 52.00, 12, 1),
129      (23, 'Travel Essentials', 'Shampoo and Conditioner', 30.00, 2, 4),
130      (234, 'Smooth Skin Collection', 'Skincare Products - Cleanser, Toner, Moisturiser', 45.00, 34, 1),
131      (235, 'Super Deluxe Collection', 'Shower Collection', 35.00, 1, 2);
132
133  -- Orders
134  • INSERT INTO Order_Data VALUES
135      (1, 3, 'Credit Card', 'Email', 'N'),
136      (2, 4, 'Cash', 'Party', 'Y'),
137      (3, 2, 'Cheque', 'Party', 'N'),
138      (4, 3, 'Cash', 'Email', 'N');
139
140  -- Order Tracking
141  • INSERT INTO Order_Tracking VALUES
142      (1, 1, 123, 25, 1300),
143      (2, 2, 23, 1, 30),
144      (3, 3, 235, 1, 35),
145      (4, 4, 23, 5, 150);
146
147  -- Delivery Information
148  • INSERT INTO Delivery_Information VALUES
149      (1, 12, 1450, 'Yes'),
150      (2, 2, 30, 'No'),
151      (3, 3, 35, 'Yes');
152
153

```

Figure 20: Data Insertion Query

Displaying First and Last Name and Displaying Alphabetically

Query:

```
SELECT Name FROM Customer_Details ORDER BY Name ASC;
```

Output:

Name
Anish Kc
Anish Singh
Jenisha Sharma
Krish Rai
Kritika Thapa
Om Thapa
Raj Joshi

The SELECT query helps to display the data or only a portion of data.

Displaying Orders by Date(Descending)

Query:

```
SELECT Order_Data.OrderRef, Order_Data.CustomerRef, Party_Data.Date
FROM Order_Data
JOIN Party_Data ON Order_Data.CustomerRef = Party_Data.HostRef
ORDER BY Party_Data.Date DESC;
```

Output:

	OrderRef	CustomerRef	Date	
	8	67	2018-04-25	
	7	34	2018-04-08	
	6	14	2018-03-24	

JOIN helps to connect records and data between two or more tables.

Counting all orders where payment method is 'Cash'

Query:

```
SELECT COUNT(*) AS CASH_PAYMENT
FROM Order_Data
WHERE Payment_Method = 'Cash';
```

Output:

CASH_PAYMENT
4

Displaying Party Orders where quantity is greater than 25

Query:

```
SELECT Order_Data.OrderRef, Party_Data.Date, Order_Tracking.Qty
FROM Order_Data
JOIN Party_Data ON Order_Data.CustomerRef = Party_Data.HostRef
JOIN Order_Tracking ON Order_Data.OrderRef = Order_Tracking.OrderRef
WHERE Order_Data.Type = 'Party' AND Order_Tracking.Qty > 25;
```


Output:

OrderRef	Date	Qty
5	2018-01-12	30
6	2018-03-24	28
7	2018-04-08	40

Updating email addresses for suppliers

Query:

```
UPDATE Supplier_Data
SET Email = CONCAT(LOWER(Name), '-supplier@', LOWER(Name), '.net');
```

Output:

SupplierID	Name	Email	Tel_Number
1	CDE	cde-supplier@cde.net	1252776789
2	GHI	ghi-supplier@ghi.net	1200467876
12	ABC	abc-supplier@abc.net	1282456762
34	DEF	def-supplier@def.net	1705674890
NULL	NULL	NULL	NULL

The update query helps to modify the data.

Adding a 'Healthy Eating' category

Query:

```
INSERT INTO Category_Data (CategoryID, Category_Name, Comments)
VALUES (5, 'Healthy Eating', "This is healthy eating category");
```

Output:

CategoryID	Category_Name	Comments
1	Skincare	Summer Season
2	Bodycare	Summer Range
3	Makeup	All minerals
4	Haircare	All products
5	Healthy Eating	This is healthy eating category
NULL	NULL	NULL

Updating customer type for Krish Rai

Query:

```
UPDATE Customer_Details
SET Customer_Type = 'Ambassador'
WHERE Name = 'Krish Rai';
```

Output:

SN	Name	Address	Number	Email	Customer_Type
2	Anish Kc	Chitwan	9844532468	anish@gmail.com	Customer
3	Om Thapa	Pokhara	9845431792	om12@cmail.com	Ambassador
4	Raj Joshi	Butwal	9845124793	raj@smail.com	Customer
12	Krish Rai	Jhapa	9845641792	krish@hmail.com	Ambassador
14	Jenisha Sharma	Kathmandu	9845112444	jenisha@xmail.com	Host
34	Kritika Thapa	Biratnagar	9845443322	kritika@yahoo.com	Host
67	Anish Singh	Lalitpur	9845000011	anish@singh.com	Host

Deleting the product 'Smooth Skin Collection'

Query:

```
DELETE FROM Product_Data
WHERE name = 'Smooth Skin Collection';
```

Output:

ProductCode	name	Description	Price	SupplierID	CategoryID
23	Travel Essentials	Shampoo and Conditioner	30	2	4
123	ABC + free eye refresh	Cleanser- Natural Ingredients	52	12	1
235	Super Deluxe Collection	Shower Collection	35	1	2
NULL	NULL	NULL	NULL	NULL	NULL

Task 3: Data Visualization

Visuals like pie chart, line graph, hist graph are made using Numpy, Pandas and Matplotlib.

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	7-Jan-18	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	15-Jan-18	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	1-Aug-18	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	8-Jun-18	Varies with device	4.2 and up
4	Pixel Draw – Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	20-Jun-18	1.1	4.4 and up

Figure 21: Dataset Preview

Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Figure 22: Correct Import of Packages

```
df = pd.read_csv("googleplaystore(in).csv")
```

Figure 23: Reading the dataset

```
df = df[df['Category'].notnull()]
df['Rating'] = pd.to_numeric(df['Rating'], errors='coerce')
df['Reviews'] = pd.to_numeric(df['Reviews'], errors='coerce')

df['Installs_clean'] = df['Installs'].str.replace('[+,]', '', regex=True)
df['Installs_clean'] = pd.to_numeric(df['Installs_clean'], errors='coerce')

def convert_size(x):
    if pd.isnull(x) or x == 'Varies with device':
        return np.nan
    if 'M' in x:
        return float(x.replace('M', ''))
    if 'k' in x:
        return float(x.replace('k', ''))/1024
    return np.nan
df['Size_MB'] = df['Size'].apply(convert_size)

df['Price_clean'] = df['Price'].replace(['$'], '', regex=True)
df['Price_clean'] = pd.to_numeric(df['Price_clean'], errors='coerce').fillna(0)

df['Rating'] = df['Rating'].fillna(df['Rating'].median())
df['Installs_clean'] = df['Installs_clean'].fillna(0)
```

Figure 24: Data Cleaning

Data cleaning involves removing missing category, removing special characters, data type conversions, filling missing values ([Guide to Data Cleaning, n.d.](#)).

```
line_data = df.groupby('Installs_clean')['Rating'].mean().reset_index()
line_data = line_data[line_data['Installs_clean'] > 0].sort_values('Installs_clean')

plt.figure(figsize=(10,6))
plt.plot(line_data['Installs_clean'], line_data['Rating'], marker='o', linestyle='--')
plt.xscale('log')
plt.title('Average App Rating vs. Number of Installs')
plt.xlabel('Number of Installs (log scale)')
plt.ylabel('Average Rating')
plt.grid(True)
plt.show()
```

Figure 25: Code for Line Graph

```
top_reviewed = df[['App', 'Reviews']].dropna().sort_values('Reviews', ascending=False).head(10)

plt.figure(figsize=(10,6))
plt.barh(top_reviewed['App'], top_reviewed['Reviews'], color='purple')
plt.xlabel('Number of Reviews')
plt.title('Top 10 Most Reviewed Apps')
plt.gca().invert_yaxis()
plt.show()
```

Figure 26: Code for Bar Graph

```
sizes = df['Size_MB'].dropna()
plt.figure(figsize=(10,6))
plt.hist(sizes[sizes < 100], bins=30, color='skyblue', edgecolor='black')
plt.title('Distribution of App Sizes (<100MB)')
plt.xlabel('App Size (MB)')
plt.ylabel('Number of Apps')
plt.show()
```

Figure 27: Code for Hist Graph

```
rating_counts = df['Content Rating'].value_counts()

plt.figure(figsize=(8,8))
plt.pie(
    rating_counts.values,
    labels=rating_counts.index,
    autopct='%1.1f%%',
    startangle=120
)
plt.title("Distribution of Apps by Content Rating")
plt.show()
```

Figure 28: Code for Bar Graph

Visuals Output

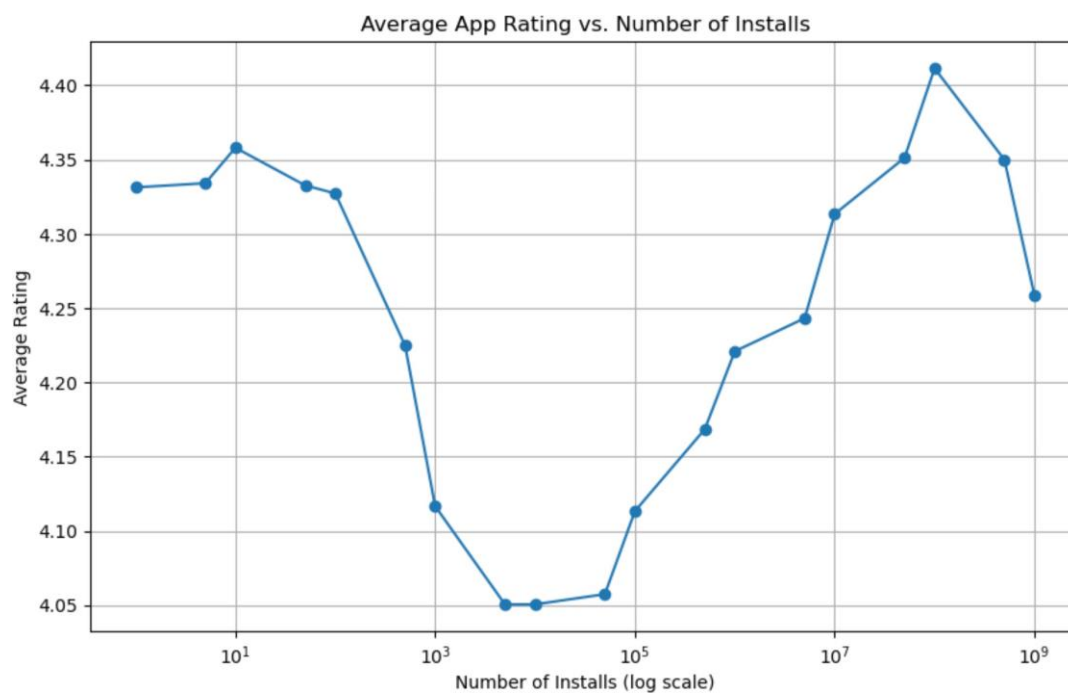


Figure 29: Line Graph

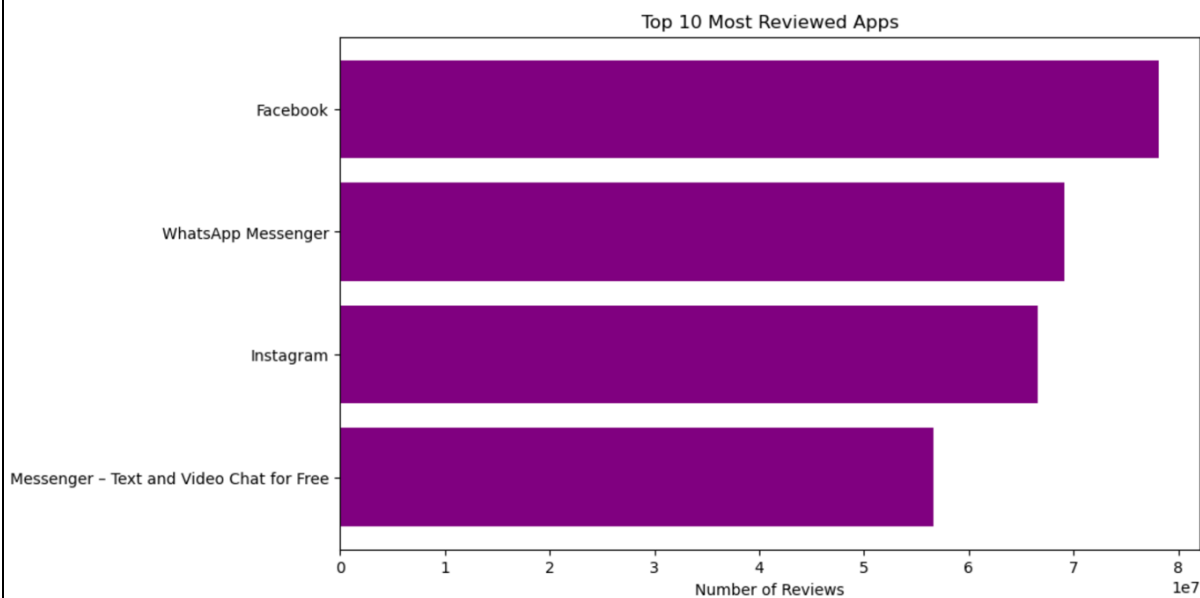


Figure 30: Bar Graph

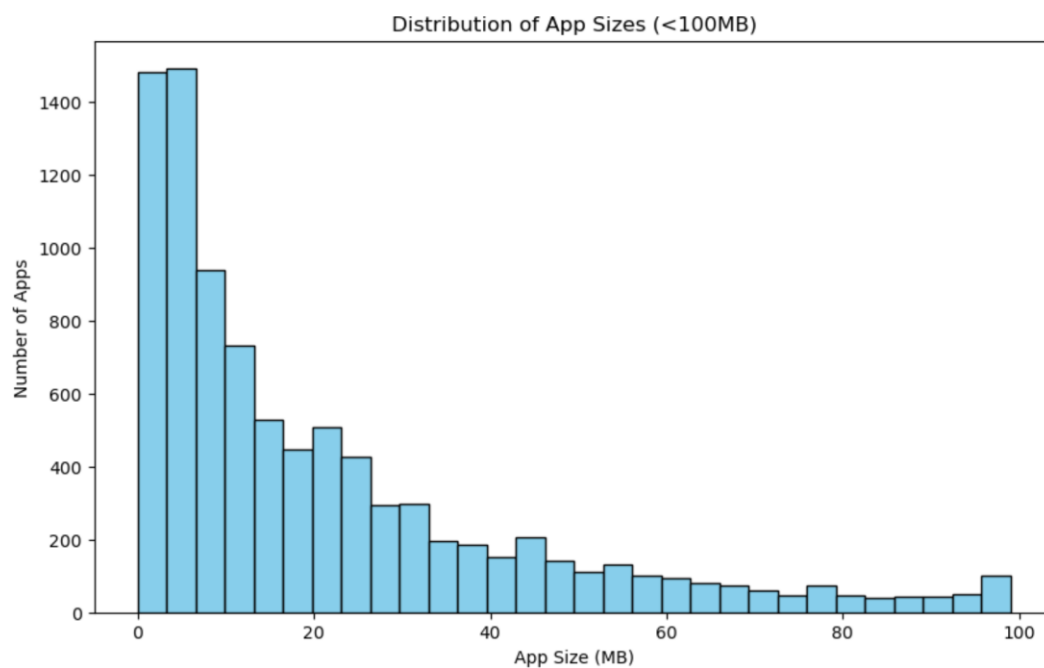


Figure 31: Hist Graph

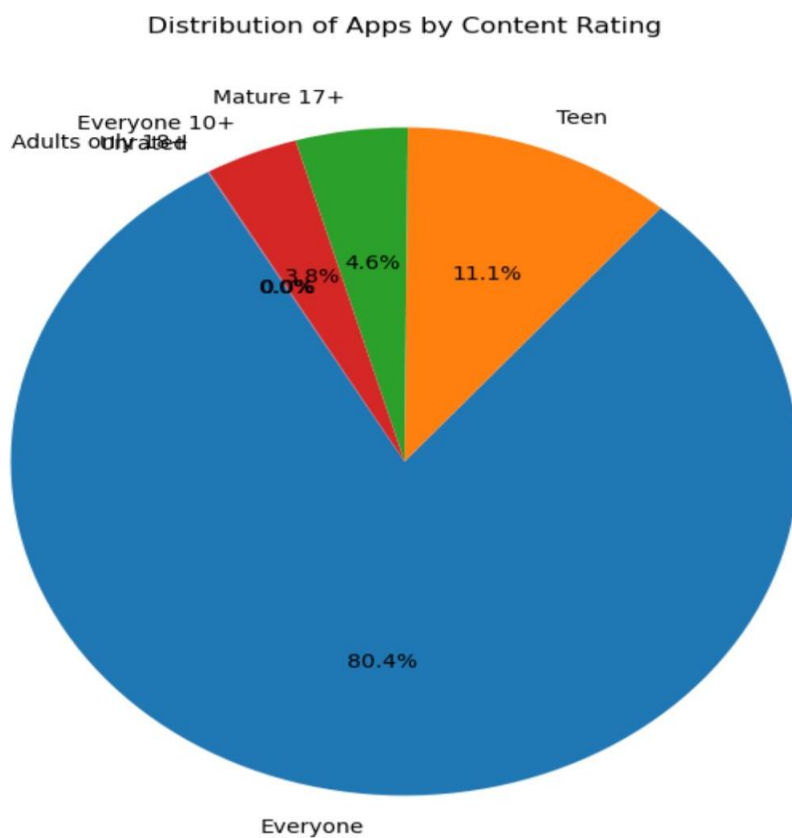


Figure 32: Pie Chart

Data Analysis

Indeed, Facebook is the most reviewed app, according to the above bar graph, average app rating remains consistent, and that most apps are designed to be light.

Conclusion

A digital transformation for records can be fully achieved via practices of normalization, Entity-Relationship Diagrams, SQL Query, and Data Visualization. The final database system enhances secured records, and reduces redundancy.

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